

FUNDAMENTALS OF COMBUSTION

PROFESSIONAL ELECTIVE – V

VIII Semester										
Course Code		Category		Hours / Week			Credits	Maximum Marks		
A6AE53		PCC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		INTRODUCTION								
Introduction to combustion, Applications of combustion, Types of fuel and oxidizers, Characterization of fuel Various combustion mode, Scope of combustion										
UNIT-II		THERMODYNAMICS OF COMBUSTION AND CHEMISTRY OF COMBUSTION								
Thermodynamics properties, Laws of thermodynamics, Stoichiometry, Thermo chemistry, adiabatic temperature, chemical equilibrium. Basic Reaction Kinetics, Elementary reactions, Chain reactions, Multistep reactions, simplification of reaction mechanism, Global kinetics.										
UNIT-III		CHEMISTRY OF COMBUSTION AND PHYSICS OF COMBUSTION								
Basic Reaction Kinetics, Elementary reactions, Chain reactions, Multistep reactions, simplification of reaction mechanism, Global kinetics. Fundamental laws of transport phenomena, Conservations Equations, Transport in Turbulent Flow										
UNIT-IV		PREMIXED FLAME AND DIFFUSION FLAME								
One dimensional combustion wave, Laminar premixed flame, Burning velocity measurement methods, Effects of chemical and physical variables on Burning velocity, Flame extinction, Ignition, Flame stabilizations, Turbulent Premixed flame. Gaseous Jet diffusion flame, Liquid fuel combustion, Atomization, Spray Combustion, Solid fuel combustion.										
UNIT-V		COMBUSTION AND ENVIRONMENT								
Atmosphere, Chemical Emission from combustion, Quantification of emission, Emission control methods.										
Text Books:										
1. D.P. Mishra, Fundamentals of Combustion, Prentice Hall of India, New Delhi, 2008. 2. Kuo K.K. "Principles of Combustion" John Wiley and Sons, 2005. 3. Strehlow R A., "Fundamentals of combustion" McGraw Hill Book Company, 1984										
Reference Books:										
1. Warnatz J, Combustion: Physical and Chemical Fundamentals, Modelling and Simulation, Experiments, Pollutant Formation, Springer Verlag, 1996. 2. Mishra D P, Fundamental of Combustion, PHI Learning, 2008.										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. Explain the thermodynamics of combustion and analyse the composition of various types of fuels and their properties. 2. Discuss the fundamental thermodynamics of combustion and basic kinetics of various reactions 3. Discuss the fundamental physical and chemical principles of various combustion phenomena independent of an application 4. Address premixed flame and diffusion by extending the earlier-gained knowledge of thermodynamics. 5. Make quantitative and qualitative estimates of characteristics of various combustion processes and its impact on environment										